TrackVision: Identification and Analysis of Player Positions and Shuttlecock Trajectory Using YOLO and TrackNet Models

Abstract— **Automated analysis of player positioning and shuttlecock dynamics is essential for performance evaluation, tactical optimization, and automated officiating in badminton. However, accurate analysis from monocular broadcast videos remains challenging due to rapid shuttlecock motion, severe motion blur, player occlusions, and perspective distortion. This paper presents an integrated deep learning framework for real-time badminton match analysis using a single-camera broadcast feed. The proposed framework combines three specialized modules: a ResNet-50-based keypoint regression network for badminton court localization and homography-based perspective transformation, a YOLOv8 object detection model for robust player detection, and a TrackNetV2-inspired spatio-temporal network for high-speed shuttlecock tracking. To improve temporal consistency and reduce detection errors, a post-processing pipeline incorporating Kalman filter-based trajectory smoothing and velocity-thresholded outlier removal is introduced. Experimental results demonstrate strong performance across all components, achieving a Percentage of Correct Keypoints (PCK@0.05) exceeding 92% for court keypoint detection and a mean Average Precision (mAP@0.5) of 93.6% for player detection, while maintaining robust shuttlecock tracking under fast-motion and partial occlusion scenarios. By transforming image-space detections into real-world court coordinates, the proposed framework enables advanced tactical analytics, including player movement trajectory analysis, court zone occupation heatmaps, shuttlecock trajectory visualization, and rally structure analysis, with the use of standard monocular broadcast footage without requiring specialized sensors or multi-camera systems.**

Keywords—Badminton, Sports Analytics, Computer Vision, Homography Transformation, Trajectory Reconstruction.

I. INTRODUCTION

Artificial Intelligence (AI) has fundamentally transformed modern sports analytics by enabling automated performance evaluation, dynamic tactical analysis, and objective, real-time decision-making. Over recent years, deep learning-based visual tracking frameworks have vastly improved the accuracy and efficiency of object detection, trajectory estimation, and multi-agent spatial tracking across diverse athletic disciplines [1], [2]. From automated offside detection in football to real-time player positioning in basketball, vision-based approaches have largely supplanted tedious manual video tagging.

However, applying computer vision to court-based racket sports—and badminton in particular—presents a unique set of technical challenges. Badminton is recognized as one of the fastest racket sports globally, characterized by rapid player direction changes, extreme motion blur, complex camera angles, and frequent physical occlusions. Furthermore, tracking the shuttlecock is exceptionally difficult due to its unique aerodynamic properties as described in [3], non-linear high-velocity trajectories (frequently exceeding 300 km/h), and minimal spatial pixel footprint in standard single-camera broadcast clips.

Traditional tracking paradigms fail to adequately address these challenges in routine athletic settings. Manual post-match video analysis is labor-intensive, subjective, and prone to human error, making rapid tactical feedback during tournaments impractical. Conversely, specialized hardware-based solutions, such as wearable inertial sensors or multi-camera stereoscopic optical arrays (e.g., Hawk-Eye systems), require substantial financial investments, complex physical installation, and dedicated venue infrastructure. Consequently, these systems remain inaccessible for lower-tier competitions, amateur training setups, and routine coaching workflows.

Recent advances in deep learning offer viable pathways toward lightweight, camera-only tracking solutions using standard monocular broadcast video feeds [1], [4], [5]. Deep convolutional neural networks (CNNs), particularly real-time single-stage object detectors like You Only Look Once (YOLO) architectures, have demonstrated efficiency in localizing players amidst dynamic court backgrounds [1]. Concurrently, heatmap-based deep tracking networks like TrackNet and TrackNetV2 have shown strong capabilities in localizing small, fast-moving objects by analyzing continuous temporal frame sequences rather than isolated static frames [4], [5].

Despite these developments, combining isolated tracking modules into a unified, metric-accurate analytics system remains problematic. Fluctuating lighting conditions, severe motion blur during racket strikes, low contrast between the white shuttlecock and court lines, and temporary target occlusions continuously degrade standalone network predictions. Moreover, raw pixel coordinates must be mapped precisely into real-world physical space to derive meaningful tactical metrics such as player recovery speeds, court coverage heatmaps, and shuttlecock velocity vectors.

To resolve these limitations, this study presents a fully integrated, multi-stage computer vision framework that unifies player detection, shuttlecock tracking, and spatial court registration into a single metric coordinate space. The framework incorporates a custom ResNet-50 regression model to automatically localize court corner keypoints for computing a robust homography matrix H, translating 2D image coordinates directly into 3D metric court space. This is coupled with a YOLOv8 detector optimized to maintain high precision under heavy motion blur while filtering out non-player personnel, alongside a TrackNetV2 architecture trained on continuous three-frame heatmap sequences for high-speed shuttlecock localization. Finally, a sequential post-processing pipeline combining outlier rejection, a 2D Linear Kalman Filter, and Savitzky-Golay polynomial smoothing is employed to reconstruct continuous trajectories through missing frames. By delivering an efficient, scalable, and non-intrusive solution operating on standard single-camera footage, the proposed framework aims to empower sports analysts, coaches, and athletes with automated tactical intelligence without requiring expensive physical infrastructure.

II. Related Work

A. Player Detection and Tracking in Sports Video

Early automated visual tracking systems [6], and spatio-temporal modeling work in [7], primarily targeted court sports by exploiting strict boundary constraints before deep learning models became prominent. The detection of players is dominated by single-stage detectors as they perform detection as a single regression pass over the whole frame, whereas a proposal-and-classify pipeline would not be practical for real-time inference [8]. Still, most of the reported systems are for team sports. Tanapatpiboon et al. compared the YOLOv5 variants on soccer video, where the transformer prediction head showed the best performance. They then projected the results to a bird's-eye view to generate positional heatmaps [9]. Naik and Hashmi combined YOLOv3 with SORT to get 93.7% multiple-object tracking accuracy, but performance dropped under severe occlusion [10]. Multi-camera methods directly attack this weakness, as in the extended Cascade Mask R-CNN with jersey-number recognition of Zhang et al. [11]. Each assumes conditions badminton does not offer: a large playing area, comparatively smooth motion, and in the last case several calibrated views.

B. Court Registration and Homography Estimation

Raw pixel positions have no physical meaning for spatial analysis, and it is a prerequisite to map image coordinates to a metric court plane. Early approaches fitted a homography per frame from the detected lines and corners [12]. Hu et al. presented landmarks like the free-throw line for basketball, but the calibration per frame was still inconsistent under occlusion [13]. Wen et al. instead stitched a panoramic court reconstruction by KLT feature tracking over a whole clip, absorbing occlusion and illumination drift [14]. Chu et al. replaced sparse keypoints by a uniform grid and learned keypoint-conditioned dynamic filters over a ResNet encoder [5]. Their finding that textureless, repetitive surfaces defeat generic convolution transfers directly to badminton, yet all four were built around soccer or basketball geometry.

C. Shuttlecock Tracking and Badminton Analytics

The shuttle path shows the rally structure and type of shot, and therefore its recovery is central to performance analysis. Huang et al. proposed TrackNet, a heatmap network of VGG-16 encoder and deconvolutional decoder, which takes consecutive frames so that the motion continuity can compensate for the blur [4]. The 98.5% F1 headline on tennis came from a single final. Under ten-fold cross-validation this fell to 84.3% and on badminton to 68.7%. That spread is the clearest evidence that the shuttlecock is the harder target. Liu and Wang closed the gap with MonoTrack by combining a lightweight TrackNetV2 variant, a GRU-based hit-segmentation network achieving roughly 90% accuracy, and physics-informed optimization to recover 3D trajectories from one view [15]. Weeratunga et al. built zone-based transition graphs of player movement using a dynamic window for segment assignment, reporting above 85% accuracy against analyst notation [16], but from proprietary annotations on narrow-field elite footage, limiting reproducibility. Previous 2-D seriate image analysis required manual indexing [17]. Badminton lags behind tennis and football in annotated resources, according to racket-sport science reviews [18], [19].

Progress has been real but compartmentalized that detection, registration and trajectory reconstruction have developed mostly as separate problems and mostly outside badminton. The most similar system is MonoTrack, but its target is the shuttle and the hit sequence, with the player pose as a cue for hit attribution rather than a reported spatial output. Three gaps follow. Badminton lacks a large public annotated datasets, shuttlecock detection under fast motion and blur remains unreliable, and no published single-camera pipeline reports court mapping, player position and shuttlecock trajectory together in one coordinate frame.

III. Methodology

In the proposed approach, a single camera broadcast footage is transformed into structured spatial data through court registration, multi object detection, and trajectory reconstruction. Video frames are pre-processed, processed by detection modules, and projected onto standardized court coordinates for tactical analysis.

A. System Overview

The pipeline (Figure 1) operates through four stages. A ResNet-50 regressor detects four court corner keypoints to compute homography matrix H, establishing a real-world spatial reference ($6.1\text{m} \times 13.4\text{m}$). A YOLOv8 detector localizes players frame by frame, while a TrackNetV2 based model processes temporal frame batches (3 consecutive frames) to generate heatmaps for shuttlecock localization. Detected pixel coordinates are projected using H, and the shuttlecock trajectory is refined with Kalman filtering and Savitzky-Golay smoothing. The system outputs JSON files containing player paths and smoothed trajectories.

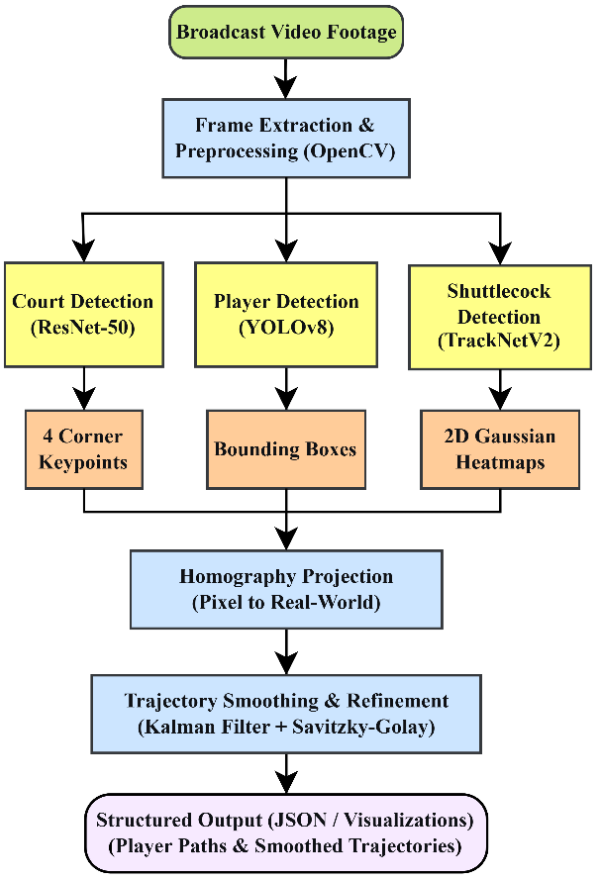


Figure 1: Architecture of the proposed badminton match analysis system illustrating the four core modules and data flow

B. *Dataset Preparation and Augmentation*

Three datasets were assembled from Olympic singles matches. Frames were annotated using Roboflow. For court detection, four corner keypoints were labeled in COCO JSON format. For players, bounding boxes included partially occluded instances. For shuttlecock tracking, Gaussian heatmaps centered on the shuttlecock were generated. Augmentation applied: horizontal flipping, rotation, brightness variation, and motion blur simulation. TABLE 1 summarizes the configurations. Additionally, to mitigate image noise and variable court lighting conditions in broadcast streams, specialized visual enhancement concepts inspired by [20] were evaluated prior to feature extraction.

TABLE 1: SUMMARY OF DATASET AND TRAINING CONFIGURATIONS

Component	Architecture	Output Type	Loss Function	Epochs
Court Detection	ResNet-50	4 Keypoints (x,y)	MSE	40
Player Detection	YOLOv8	Bounding Boxes	Compound (CIoU + BCE)	100
Shuttlecock	TrackNetV2	Heatmap (Gaussian)	WBCE	30

C. *Detection Model Architectures*

- **Court Keypoint Detection (ResNet-50):** ImageNet pre-trained ResNet-50 was adapted for regression. The final fully connected layer was replaced with a linear layer (8 outputs). Optimized using MSE loss with Adam (1×10^{-4}).
- **Player Detection (YOLOv8):** YOLOv8 initialized with COCO pretrained weights predicts class probabilities, bounding boxes, and objectness scores. Compound loss: CIoU (localization), binary cross entropy (class), and focal loss (confidence).
- **Shuttlecock Detection (TrackNetV2):** A TrackNetV2 based U-Net encoder decoder with skip connections processes three consecutive frames. Outputs a 2D heatmap modeling the shuttlecock as a Gaussian blob. Weighted Binary Cross-Entropy (WBCE) loss addresses foreground background class imbalance.

D. *Spatial Mapping and Trajectory Smoothing*

Detected keypoints were used to compute homography matrix H via Direct Linear Transform, mapping image points (x_i, y_i) to real-world court coordinates (X_w, Y_w):

$$\begin{bmatrix} X_w \\ Y_w \\ 1 \end{bmatrix} \sim H \begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix} \quad (1)$$

where H is a 3×3 matrix estimated using OpenCV's RANSAC-based findHomography. To reduce jitter and compensate for occlusions, a linear Kalman filter (position and velocity states) was applied to shuttlecock trajectories, followed by a Savitzky-Golay filter (window size 15, polynomial order 3). Outliers exceeding frame-to-frame displacement thresholds were removed.

E. *Implementation Details*

All models were implemented in Python 3.9 using PyTorch 2.0 and OpenCV 4.8. Training and inference were performed on an NVIDIA RTX 3090 GPU (24GB VRAM). The integrated pipeline achieves real time processing at over 30 FPS on broadcast resolution video.

IV. RESULTS AND DISCUSION

A. *Quantitative Performance of System Components*

Each module was evaluated separately on held-out frames drawn from the annotated Olympic singles footage described in Section III.

1) Court Keypoint Localization: The ResNet-50 keypoint regressor converged smoothly over 40 training epochs. As shown in the training loss curve in Figure 2, the Mean Squared Error (MSE) loss fell rapidly from an initial baseline of approximately 1.6×10^5 to near zero by epoch 25. This steady loss reduction yields a final Root Mean Squared Error (RMSE) of under 5 pixels in standard $1080p$ frame space. On the PCK@0.05 metric, which measures the percentage of keypoints predicted within a 5% distance threshold relative to court dimensions, the model exceeded 92% accuracy. Sub-five-pixel corner accuracy ensures that the computed homography matrix H introduces minimal geometric distortion during spatial coordinate projection.

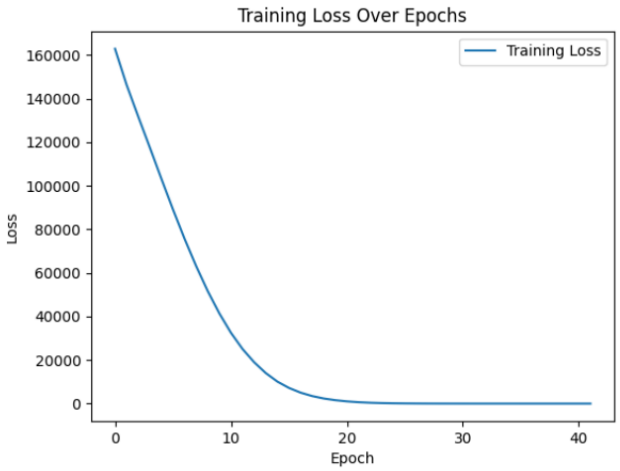


Figure 2: Training Loss Curve by Epoch for the Badminton Court Keypoint Detection

2) Player Detection and Identification: The YOLOv8 detector was trained for 100 epochs. Figure 3 presents the box regression, classification, and distribution focal losses for both the training and validation splits, together with precision, recall, and mean Average Precision (mAP). All six loss curves decline monotonically, and the validation curves track the training curves throughout without diverging, confirming that the model did not overfit. Detection reached a mAP@0.5 of 93.6%, with a mAP@0.5:0.95 of approximately 78%. Maximum precision approached 1.00 at a confidence threshold near 0.81, which was adopted at inference to exclude non-player personnel (such as line judges, coaches, and spectators) from court positioning statistics.

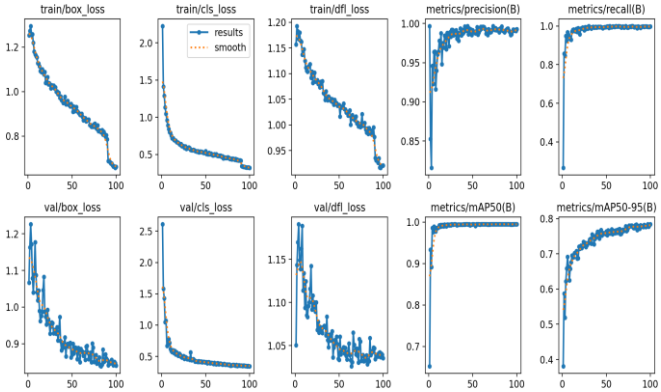


Figure 3: YOLOv8 training and validation loss curves alongside precision, recall, and mAP metrics across 100 epochs.

3) Shuttlecock Tracking Fidelity: TrackNetV2 was trained for 30 epochs on three-frame inputs using Weighted Binary Cross-Entropy loss. As Figure 4 illustrates, detection accuracy passed 80% by epoch 5 and stabilized at 88.49%, achieving a test precision of 98.36% and a recall of 89.42%. The higher precision relative to recall indicates that the network misses' shuttlecock more often than it invents false detections. For spatial trajectory reconstruction, this represents the preferred failure mode, as a dropped frame can be seamlessly interpolated during trajectory smoothing, whereas a false positive introduces a spurious point that corrupts the flight path.

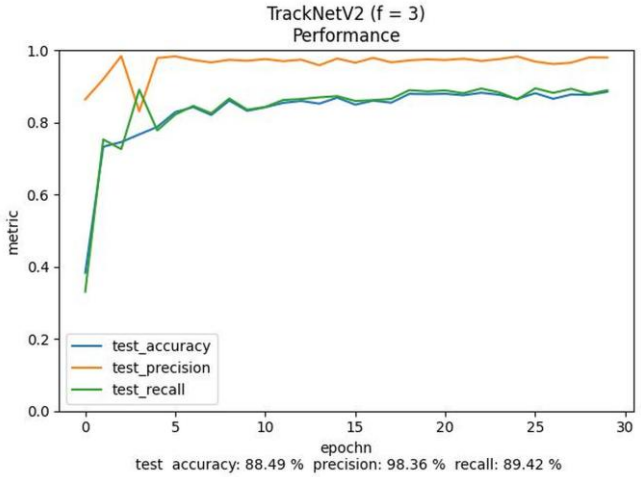


Figure 4:TrackNetV2 test accuracy, precision, and recall across training epochs

TABLE 2: QUANTITATIVE EVALUATION OF SYSTEM MODULES

Component	Architecture	Primary Metric	Reported Metrics
Court Detection	ResNet-50	PCK@0.05	Accuracy > 92%; RMSE below 5 px
Player Detection	YOLOv8	mAP@0.5	mAP@0.5 = 93.6%; mAP@0.5:0.95 = 78%
Shuttlecock	TrackNetV2	Accuracy	Accuracy 88.49%; Precision 98.36%; Recall 89.42%

B. *Qualitative Detection Results*

Figure 5 demonstrates all three modules operating simultaneously on a single broadcast frame. The four numbered markers represent the court corner keypoints regressed by ResNet-50 used to calculate H. The two bounding boxes correspond to YOLOv8 player detections retained after proximity filtering filtered out non-player personnel. The shuttlecock label marks the peak detection heatmap from TrackNetV2. Successful simultaneous inference across all three modules visually confirms the quantitative metrics reported above.

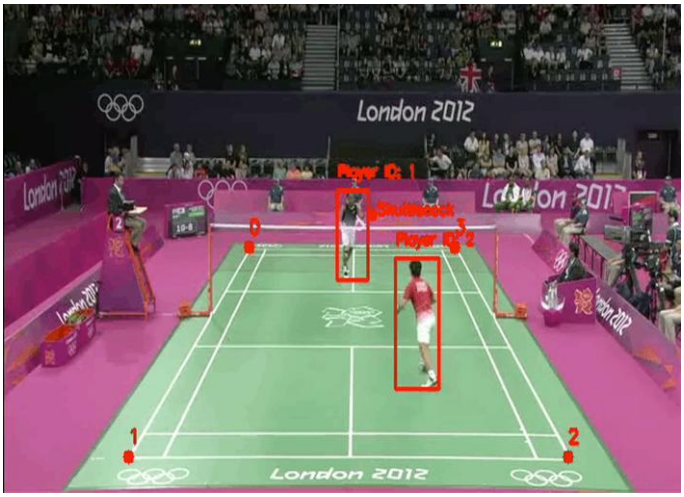


Figure 5: Combined system output on a single broadcast frame showing court corner keypoints, player bounding boxes, and shuttlecock detection

C. Trajectory Refinement and Spatial Mapping

Detected pixel coordinates were projected into metric court space (6.1 m x 13.4 m) through the homography matrix H and then refined in three stages. First, frame-to-frame displacement was computed and detections implying physically implausible speeds were discarded as outliers, since a shuttlecock cannot traverse the court between consecutive frames. Second, a linear Kalman filter modelled the shuttlecock state as position and velocity, with the state vector $x = [x, y, v_x, v_y]^T$ evolving under a constant-velocity transition model. The filter was initialised at the first valid detection and updated frame by frame; when no detection was available for a frame, the predicted state was carried forward, which is what allows the reconstructed track to survive the missed detections implied by the 89.42% recall reported above. Third, a Savitzky-Golay filter fitted a low-degree polynomial over a sliding window of the Kalman output, suppressing residual jitter while preserving the sharp direction changes that follow a racket strike. A window length of 15 frames and a polynomial order of 3 were used, matching the configuration reported in Section III-D. An over-long window would round off exactly the trajectory features that mark a hit, so this parameter choice balances smoothness against responsiveness.

D. Comparison with Prior Work

TABLE 3 places the shuttlecock tracking module in the context of previously reported results. The comparison is indicative rather than controlled, since each system was evaluated on a different dataset and no common badminton benchmark currently exists. TrackNet reported 98.5% F1 on a single tennis final, 84.3% under ten-fold cross-validation on tennis, and 68.7% on badminton. The present accuracy of 88.49% on badminton frames therefore sits substantially above the published badminton figure for the original architecture, which is consistent both with the shuttlecock being a harder target than a ball and with TrackNetV2 being designed to close that gap. The more substantive difference is one of scope. MonoTrack recovers the shuttle and the hit sequence and uses player pose as a cue for hit attribution, whereas the present system reports player positions and shuttlecock trajectory together in a single metric court frame, addressing the integration gap identified in Section II.

TABLE 3: SHUTTLECOCK TRACKING RESULTS IN CONTEXT

System	Sport	Reported result
TrackNet [4]	Tennis	98.5% F1 (single match);84.3% F1 (10-fold)
TrackNet [4]	Badminton	68.7% F1
MonoTrack [15]	Badminton	Approx. 90% hit-segmentation accuracy
This work	Badminton	88.49% accuracy; 98.36% precision

E. Discussion and Tactical Applications

Integrating court registration, player tracking, and shuttlecock detection into a single metric frame transforms raw broadcast features into tactical athletic intelligence. Projecting bottom-center bounding boxes into standard court dimensions automates key physical metrics like distance covered, recovery speed relative to base position, and occupancy heatmaps. Concurrently, smoothed shuttlecock trajectories yield velocity profiles where speed drops and directional reversals identify exact impact timestamps. Correlating these timestamps with player positioning extracts tactical insights—including shot origin, contact position, and transit time—for automated match analysis.

F. Limitations

Despite the overall robustness of the proposed framework, four primary constraints qualify these findings. First, operating on monocular broadcast footage inherently lacks direct multi-view stereo triangulation, restricting 3D shuttlecock altitude estimation to physics-based parabolic models. Second, high-density exchanges near the net and physical player crossovers occasionally introduce brief bounding box overlaps or temporary identity swaps. Third, environmental variances in broadcast streams—such as harsh venue spotlights, floor reflections, or low frame bitrates—introduce localized pixel noise that reduces detection confidence during rapid shuttlecock movements. Finally, because the experimental evaluation relied exclusively on single-camera Olympic singles matches, the reported metrics demonstrate performance on high-quality broadcast footage, leaving doubles play, alternative broadcast angles, and amateur recording conditions subject to further empirical validation.

V. Conclution And Future Works

This study presented Track Vision, an integrated deep learning framework for real-time spatial and tactical analysis of badminton matches using monocular broadcast video. By combining a ResNet-50 landmark regressor, YOLOv8 player detector, and TrackNetV2 shuttlecock tracker, the proposed system unifies court registration, player tracking, and shuttlecock trajectory recovery into a single coordinate frame. Quantitative evaluations demonstrate strong performance across all pipeline components, achieving a Court Keypoint Regression PCK@0.05 exceeding 92%, a Player Detection mAP@0.5 of 93.6%, and Shuttlecock Detection test precision and recall of 98.36% and 89.42%, respectively. By leveraging homography mapping alongside Kalman filtering and velocity-thresholded outlier rejection, the framework maintains spatio-temporal continuity despite severe motion blur and occlusions. Operating at over 30 FPS on high-definition video, TrackVision provides an accessible, non-invasive, and cost-effective alternative to multi-camera setups for automated performance analysis.

Future work will expand the system's analytical scope, detection robustness, and practical deployment. Priority will be given to supporting doubles and mixed doubles formats by engineering multi-player tracking algorithms and team-assignment logic capable of handling four simultaneous players and complex rotational dynamics. Integrating high-resolution pose estimation frameworks, such as HRNet introduced in [21], or OpenPose will enable fine-grained biomechanical action recognition, allowing automated classification of smashes, drops, and defensive stances. To better manage extreme object velocities, future iterations will explore transformer-based backbones like DETR or Vision Transformers for long-range spatial

reasoning, alongside processing high-temporal resolution inputs (120–240 FPS) to eliminate motion blur artifacts. Finally, extending dataset diversity to include amateur and variable court conditions will improve generalization, while introducing automated rally segmentation and tactical summarization modules will deliver actionable insights for grassroots coaching and sports broadcasting.

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